

Justify your answers and show your work. All questions are of equal value.

1. Write the infinite series, $0.3 + 0.03 + 0.003 + \cdots$, using sigma notation (i.e. using Σ).

$$\sum_{k=1}^{\infty} \frac{3}{10^k}$$

2. Write the first 3 terms of the sequence of partial sums of the series $\sum_{k=1}^{\infty} \frac{3^k}{2}$.

$$\sum_{k=1}^3 \frac{3^k}{2} = \frac{3}{2} + \frac{9}{2} + \frac{27}{2}$$

Evaluate the geometric series or state that it diverges.

3. $\sum_{k=0}^{\infty} \frac{2^k}{7^k}$.

$$\sum_{k=0}^{\infty} \frac{2^k}{7^k} = \sum_{k=0}^{\infty} \left(\frac{2}{7}\right)^k = \frac{1}{1 - 2/7} = \frac{7}{5} \text{ since } r = \frac{2}{7} \text{ and } a = 1.$$

4. $\sum_{m=2}^{\infty} \left(\frac{1}{3}\right)^m$.

$$\sum_{m=2}^{\infty} \left(\frac{1}{3}\right)^m = \sum_{m=0}^{\infty} \left(\frac{1}{3}\right)^{m+2} = \sum_{m=2}^{\infty} \left(\frac{1}{3}\right)^2 \left(\frac{1}{3}\right)^m = \frac{1/9}{1 - 1/3} = \frac{1}{6} \text{ since } r = \frac{1}{3} \text{ and } a = \frac{1}{9}. \text{ Note that you can also just write out a few terms to determine the } a \text{ and } r.$$

Find the limit of the following sequences or determine that the limit does not exist.

5. $\lim_{n \rightarrow \infty} \left\{ \frac{n^4 - 1}{n^4 + 10} \right\}$.

$$\frac{n^4 - 1}{n^4 + 10} \frac{1/n^4}{1/n^4} = \frac{1 - 1/n^4}{1 + 10/n^4} \Rightarrow \lim_{n \rightarrow \infty} \left\{ \frac{n^4 - 1}{n^4 + 10} \right\} = \lim_{n \rightarrow \infty} \left\{ \frac{1 - 1/n^4}{1 + 10/n^4} \right\} = 1$$

6. $\lim_{n \rightarrow \infty} \left\{ \frac{n^{0.01}}{\ln n} \right\}$.

$$\ln n \ll n^{0.01} \text{ so then } \lim_{n \rightarrow \infty} \left\{ \frac{n^{0.01}}{\ln n} \right\} = \infty.$$

Use the Divergence Test to determine whether the following series diverge, or state that the test is inconclusive (use your results from above).

7. $\sum_{k=1}^{\infty} \frac{k^4 - 1}{k^4 + 10}$.

$$\text{From above, } \lim_{k \rightarrow \infty} \left\{ \frac{k^4 - 1}{k^4 + 10} \right\} \neq 0, \text{ so by the Divergence theorem } \sum_{k=1}^{\infty} \frac{k^4 - 1}{k^4 + 10} \text{ diverges.}$$

8. $\sum_{k=1}^{\infty} \frac{k^{0.01}}{\ln k}$.

$$\text{From above } \lim_{n \rightarrow \infty} \left\{ \frac{n^{0.01}}{\ln n} \right\} \neq 0 \text{ so by the Divergence theorem } \sum_{k=1}^{\infty} \frac{k^{0.01}}{\ln k} \text{ diverges.}$$

Using the Squeeze Theorem, find the limit of the following sequence or determine that the limit does not exist.

9. $\lim_{n \rightarrow \infty} \left\{ \frac{\sin^2 n}{2^n} \right\}.$

$$-1 \leq \sin x \leq 1 \Rightarrow 0 \leq \sin^2 x \leq 1. \text{ So then } 0 \leq \frac{\sin^2 n}{2^n} \leq \frac{1}{2^n} \Rightarrow 0 = \lim_{n \rightarrow \infty} \{0\} \leq \lim_{n \rightarrow \infty} \left\{ \frac{\sin^2 n}{2^n} \right\} \leq \lim_{n \rightarrow \infty} \left\{ \frac{1}{2^n} \right\} = 0. \text{ So then } \lim_{n \rightarrow \infty} \left\{ \frac{\sin^2 n}{2^n} \right\} = 0 \text{ by the Squeeze theorem.}$$

Use the Integral Test to determine whether the series converge, diverge, or state why the test cannot be used. Check that the conditions of the test are satisfied.

10. $\sum_{k=1}^{\infty} \frac{k}{e^k}.$ For $f(x) = \frac{x}{e^x}$, $f'(x) = (1-x)e^{-1}$ and $\int f(x) dx = 2xe^{-1}.$

For $x \geq 1$, $f'(x) = (1-x)e^{-1} \leq 0$, $f(x) = \frac{x}{e^x} \geq 0$, and $f(x)$ is continuous. So by the integral test it diverges since,

$$\int_1^{\infty} \frac{x}{e^x} = \lim_{b \rightarrow \infty} (2xe^{-1}|_1^b) = \lim_{b \rightarrow \infty} 2xe^{-1} - 2e^{-1} = \infty - 2e^{-1} = \infty.$$

11. $\sum_{k=2}^{\infty} \frac{1}{(2x-3)^2}.$ For $f(x) = \frac{1}{(2x-3)^2}$, $f'(x) = \frac{-4}{(2x-3)^2}$ and $\int f(x) dx = \frac{-1}{2(2x-3)}.$

For $x \geq 2$, $f(x)$ is continuous, $f'(x) \leq 0$, and $f(x) \geq 0$. So by the integral test the series converges since

$$\int_2^{\infty} \frac{1}{(2x-3)^2} dx = \lim_{b \rightarrow \infty} \frac{1}{2(2x-3)} \Big|_2^b = 0 - (-1/2) = 1/2$$

12. Let $f(x) = 2 \cos(x).$

a. Find the Taylor polynomial of order $n = 3$ for $f(x)$ centered at 0.

$$f'(x) = -2 \sin(0) = 0, f''(x) = -2 \cos(0) = -2, f^{(3)}(0) = 2 \sin(x) = 0. \text{ So then } p_3(x) = 2 + 0 - x^2.$$

b. Use the Taylor polynomial from above to approximate $2 \cos(0.1).$

$$p_3(0.1) = 2 - (.1)^2 = 1.99$$

c. Compute the absolute error in the approximation assuming the exact value is given by a calculator.

$$|p_3(0.1) - 2 \cos(0.1)| \approx .00001$$